

Week 3 – Memory and Stack Test Results

Memory Tests

Test	Expected Result	Actual Result	Status
RAM READ/WRITE	Written value should be read correctly	Value read correctly	PASS
RAM Address 0	Address 0 should work correctly	Address 0 worked correctly	PASS
RAM Address 255	Address 255 should work correctly	Address 255 worked correctly	PASS
RAM 8-bit Masking	Values should be limited to 8 bits	8-bit masking worked correctly	PASS

Stack Tests

Test	Expected Result	Actual Result	Status
PUSH/POP	Pushed value should be returned by POP	Value returned correctly	PASS
LIFO	Last pushed value should come out first	LIFO worked correctly	PASS
Stack Pointer (SP)	SP should change correctly during PUSH/POP	SP worked correctly	PASS
13-bit Masking	Stack values should be limited to 13 bits	13-bit masking worked correctly	PASS
Underflow	POP from empty stack should be handled	Underflow handled correctly	PASS
Overflow	More than 8 stack values should be rejected	Overflow handled correctly	PASS
Reset	Memory and stack should return to reset state	Reset worked correctly	PASS

Overall Result

Component	Result
Memory Testing	All tests passed
Stack Testing	All tests passed
Boundary Testing	Passed
Error Testing	Passed
Reset Testing	Passed
Overall	Memory + Stack testing complete